ECONNRESET on a reliable socket and now?

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ECONNRESET and Reliability

ECONNRESET is an Errno 104:

- ERRORS: `man 2 sendmsg`
- “Connection reset by peer”

Reliablility in this talk:

- Connection Oriented Communication
- TCP and SCTP (no partial reliability)
  - Retransmit if necessary
  - Guaranteed Ordering

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Why ECONNRESET occurs?

- RFC3360 - “Inappropriate TCP Resets Considered Harmful”[1)]
  - History (reserved/no listen socket)
  - Firewalls
  - Load Balancers (ECMP[2)])
- Network reconfiguration? -> Suddenly
- TCPkill[3)], SCTP? (Heartbeat)

1) https://datatracker.ietf.org/doc/html/rfc3360
2) https://www.fastly.com/blog/building-and-scaling-fastly-network-part-2-balancing-requests
3) https://www.monkey.org/~dugsong/dsniff/ - dsniff (recommended to use Debian patches)

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Effects of ECONNRESET

Socket cannot be used anymore,
usually a reconnect is required



Losing Socket State
(different than EAGAIN)

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Show me some TCP Code

```
/* When we get a reset we do this. */
void tcp_reset(struct sock *sk, struct sk_buff *skb)
{
        ...
        /* We want the right error as BSD sees it (and indeed as we do). */
        switch (sk->sk_state) {
                ...
        default:
                sk->sk_err = ECONNRESET;
        }

        ...

        tcp_write_queue_purge(sk);
        tcp_done(sk);

        if (!sock_flag(sk, SOCK_DEAD))
                sk_error_report(sk);
}
```



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Distributed Lock Manager

- Linux Cluster Manager (Nodes, Res.)
  - Join Membership
  - Leave Membership

  } **Reliability Required**
  - Fence (STONITH) → Close Connection
- Locking Distributed Resources
  - Shared Block-Device e.g. iSCSI
  - Users: Cluster Filesystems e.g. GFS2
- Trusted Networks only*

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DLM and ECONNRESET

ECONNRESET occurs during reliability requirement phase

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DLM Error Handling

- Error triggers reconnect
- Will lose Socket State → DROP
- No Session Layer to handle it



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So why this talk?

- Difficult to say “fix your network”
- Be prepared for Error cases

Providing a generic solution to not worry about such problem and existing Applications can switch

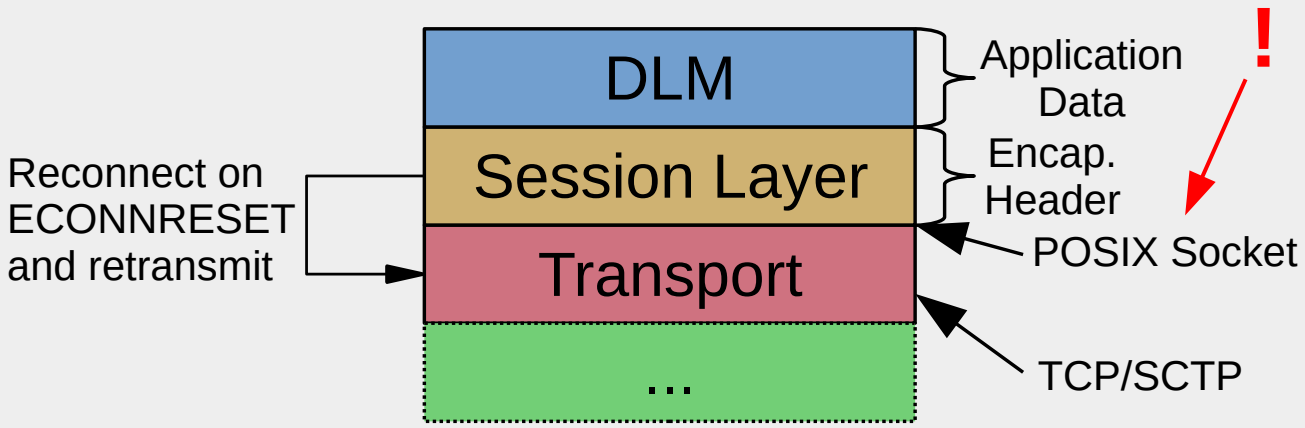
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Introduce a Session Layer

- Ideas: (~~Ignore RST~~, ~~*TCP_REPAIR*~~)
- Between any Transport and Application
- Independent from Socket Lifetime
- Inc. Sequence and Acknowledge MSG
- Keep references of sendmsg() buffers
- Resend Unacknowledged Buffers at Reconnect (Guillaume Nault)

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How DLM solved it



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How DLM solved it

- Still backwards compatible
- Close/Fence will purge everything
- Special: Half-Closed Sockets[1)]
  - Synchronize Membership Events
  - Not Supported by SCTP
  - Syscall shutdown() and ECONNRESET
  - Done on Protocol Layer not Socket

1) https://www.excentis.com/blog/tcp-half-close-cool-feature-now-broken

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Generic Solution

- Migrate the DLM Solution
- Special Socket “POSIX-Redirection”
- In-Kernel Session Layer
  - Hides ECONNRESET from User Socket
  - Hides Reconnect Process from User Socket
  - Works like DLM encap. Session Layer

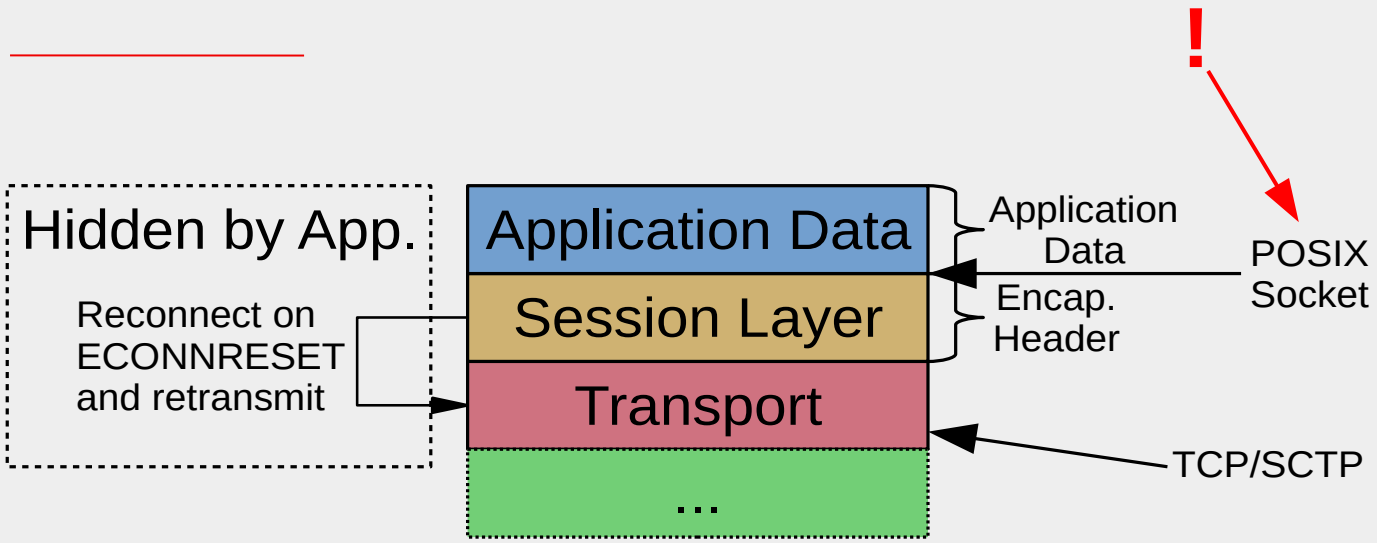
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Generic Solution

- Membership – Session Lifetime
  - Join → connect() – Gen. Unique Hash
  - Reconnect vs New Session → Hash
  - Leave → shutdown()
  - Fence → close() will purge everything
- Magic Number Header to Identify Use
- Not required by Peers (b. compatible)

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Generic Solution



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Generic Solution

- TCP Bytestream will packed into MSGs
  - User view still Stream oriented
  - Message conversion will be hidden
- Add specific Transport Layer Handling
  - Window Size Restore?
  - Zero Copy in Socket Redirection?

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Potential other Users

Follow the Checklist!

1) Join/Leave Membership
2) Reconnects on Errors
3) ?Req? Test with `tcpkill`

Let’s try CIFS!

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Potential other Users

CIFS (Client) – Samba (Server)

1) Mount/Unmount Events
2) Reconnects on Errors
3) Test with `tcpkill`

What about Windows?

Linux only

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Show me some CIFS Code

```
                ...
                rc = smb_send_kvec(server, &smb_msg, &sent);
                if (rc < 0)
                        goto unmask;
                ...
        }
        ...
unmask:
        ...

        if ((total_len > 0) && (total_len != send_length)) {
                ...
                server->tcpStatus = CifsNeedReconnect;
                ...
                trace_smb3_partial_send_reconnect(server->CurrentMid,
                                                  server->conn_id, server->hostname);
        }
```



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CIFS TCPkill Oberservation

- CIFS `tcpkill` Testcase
  - While-true loop
  - dd if=/dev/zero of=file ....
- POSIX FS: EAGAIN and ECONNRESET
- Program `dd` got stuck – still killable

Errors are side effects to ECONNRESET!

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Future Work

- Implement Generic Solution
  - Defining a Protocol
  - Half-Closed Sockets
  - Library for other OS?
- Try out CIFS with Solution
- DLM to Generic Solution?

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Conclusion

Network Devices (Firewalls or Load Balancers) can make problems

Closed Source: You need to provide the solution

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End

Thanks

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